XV Congress of the Argentine Association for Bioinformatics and Computational Biology (CA2BC)

Scientific program 11-12 November 2026 || Buenos Aires, Argentina

Wednesday, 11 November 2026	
08:00 - 09:00	***Registration***
09:00 - 09:15	***Opening Ceremony***
9:15 - 10:00	Opening Keynote Talk: **Ana Tereza Ribeiro de Vasconcelos** *(National Laboratory for Scientific Computing, Brazil)* *From genomes to knowledge: bioinformatics at national scale in Brazil*
10:00 - 10:15	***Coffee Break***
Session 1: Proteomics and Protein Structure and function	
Chair: Gustavo Parisi	
10:15 - 10:45	Invited Speaker: **Ariela Vergara Jaque** *(Facultad de Ingeniería Talca, Chile)* *Computational strategies for designing modulators of ion channel protein–protein interactions*
10:45 - 11:15	Invited Speaker: **Miguel Andrade** *(Johannes-Gutenberg University in Main, Germany)* *Sequence, context, and AI: studying the functional landscapes of intrinsically disordered regions*
11:15 - 11:55 *Short talks selected from abstracts (8'+2' questions)*	**Ornella Falduti** *(IBBM, UNLa Plata, Argentina)* *In silico prediction of (p)ppGpp alarmone binding to DksA and NodD1 in Bradyrhizobium diazoefficiens: a possible link between the stringent response and nodulation gene expression*
	Juliana Glavina *(UNSAM, BA, Argentina)* *A Novel Experimentally-derived Dataset for Structural and Sequence Analysis of Disordered Flexible Linkers*
	Exequiel Barrera *(IHEM, UNCuyo, Mendoza, Argentina)* *Simulation-guided structural modelling of amyloid-like fibrils. From isolated peptides to nanoscale assemblies*

	Galo Ballati *(UNQ, BA, Argentina)* *Molecular dynamics simulations support a dynamic RNA-recognition hotspot in the intrinsically disordered N-terminal domain of the androgen receptor*
11:55 - 12:00	***Short Break***
12:00 - 12:45	Round Table 1: **Perspectives of Artificial Intelligence in the life sciences** Chairs: Salvador Capella & Georgina Stegmayer GaborErdos (ELTE, HU) Jiri Vondrasek (ELIXIR,CZ) Ana Tereza de Vasconcelos (NLSC, BR) Henning Hermjakob (EMBL-EBI, UK) Rodrigo Diaz (UNSAM, AR)
12:45 - 14:00	***Lunch Break***
Session 2: Evolutionary, Comparative and Population Genomics	
Chair: Patricia Saragüeta	
14:00 - 14:30	Invited Speaker: ***Pierre Luisi*** *(UNC, Argentina)* *Ancient and present-day genomes in the southern cone: challenges for reconstructing population history*
14:30 - 15:00	Invited Speaker: ***Juliana Vianna*** *(UNAB, Chile)* *From reference genomes to population genomes: evolution and adaptation of South American and Antarctic vertebrates*
15:00 - 15:45 *Short talks selected from abstracts (8'+2' questions)*	**Francisco Pisciottano** *(IBYME-CONICET, Argentina)* *Assessing the Impact of Kinship on Population Genomic Analyses in the Jaguar (Panthera onca)*
	Abril Pagnutti *(UNLa Plata, Argentina)* *Protein intrinsic disorder is associated with replicon location in the multipartite prokaryotic genome of Sinorhizobium meliloti"*

	Francisco Gonzalez Espinosa *(UBA, Argentina)* *Bioinformatics applied to the study of the population dynamics of carbapenem-resistant Klebsiella pneumoniae ST258 in Buenos Aires Hospitals."*
15:45 - 16:00	***Coffee Break***
Session 3: Genomics, Transcriptomics, Epigenetics and Metagenomics	
Chair: Patricia Saragüeta	
16:00 - 16:30	Invited Speaker: ***Rodrigo Nores*** *(UNC, Argentina)* *Paleogenomic Analysis Reveals 8,000 Years of Indigenous Lineage Continuity in Human Populations of Central Argentina*
16:30 - 16:45	Invited Speaker: ***Salvador Capella Gutierrez*** *(BSC, CNS, Spain)* *ELIXIR: Ten priority areas for AI in life sciences. How do we move forward?*
16:45 - 17:00	Invited Speaker: ***Laura Kamenetzky*** *(UBA, Argentina)* *Consolidation of a long-read sequencing genomics center in southern South America*
17:00 - 17:45 *Short talks selected from abstracts (8'+2' questions)*	**Carina Fariñas** *(IBYME, BA, Argentina)* *Epigenomic and transcription factor profiles distinguish Progesterone Receptor binding sites with and without the canonical motif*
	Bruno Breggia *(sinc, UNL, Sta Fe, Argentina)* *An attention-based deep model for RNA secondary structure prediction*
	Luciana Ant *(IBYME, BA, Argentina)* *Temporal-spatial genomic regulation of the human endometrium*
	Daniela Orchanski *(UNC, Cordoba, Argentina)* *FusionAR-KB: A Multi-Layered Knowledgebase for Transcript-Specific Architecture and Clinical Interpretation of Gene Fusions in Cancer*
17:45 - 17:55	Sponsor Talk
17:55 - 19:30	Poster Session
19:30 - 20:30	***Welcome Cocktail - Art, Science and Technology Event***

Thursday, 12 November 2026	
Session 4: Biological Databases and Bioimaging	
Chair:	
09:00 - 09:30	Invited Speaker: ***Henning Hermjakob*** *(EMBL-EBI, UK):* *The complex portal: from mass spectra to structures*
09:30 - 10:00	Invited Speaker: ***Cristina Marino*** *(F. Leloir, Argentina)* *Exploring the conformational landscape of the amylome*
10:00 - 10:30 *Short talks selected from abstracts (8'+2' questions)*	**Martina Ceballos** *(UC Cordoba, Argentina)* *Computational analysis of human iris colour: from segmentation to quantitative phenotyping*
	Alexander Mulet *(UBA, BA, Argentina)* *Automatic segmentation of glioblastoma and its hidden tumor infiltration*
	M. Victoria Nugnes *(UPadova, Italy)* *Ordering the Disorder: Updates to the Intrinsically Disordered Proteins Ontology*
10:30 - 10:45	***Coffee Break***
Session 5: Artificial Intelligence and Systems Biology	
Chair: Georgina Stegmayer	
10:45 - 11:15	Invited Speaker: ***Silvio Tosatto*** *(U Padova, Italy)* *The DOME ecosystem: from recommendations to applications in Artificial Intelligence*
11:15 - 11:45	Invited Speaker: ***Juan Cortés*** *(CNRS, France)* *Generative models for the design of disordered proteins*
11:45 - 12:40 *Short talks selected from abstracts (8'+2' questions)*	**John Hancock** *(University of Ljubljana, Slovenia)* *Making Models That Matter: How to Build Trustworthy and Useful Systems Biology Models*
	Ariel Chenomoretz *(UBA, F. Leloir, BA, Argentina)* *Network organization of protein binding-site landscapes across scales*

	Leonardo Pedraza *(INTECNUS, Rio Negro, Argentina)* *From human DNA to plant regulatory regions: transfer learning with a Genomic Large Language Model*
	Santiago Britez *(F. Leloir, BA, Argentina)* *Monosemantic Decomposition of Protein Language Models via Sparse Autoencoders for Interpretable Missense Variant Classification*
	Nicolas Yanovsky *(F. Leloir, BA. Argentina)* *When perturbation queries fail: static gene and protein embeddings outperform simulated scFM perturbation responses*
12:40 - 12:45	***Short Break***
12:45 - 13:30	Round Table 2: **Leveraging research infrastructures for bioinformatics: the ELIXIR experience** Chairs: Lucía Chemes & John Hancock Jimena Rinaldi (CEBEM & UNSAM, AR) Salvador Capella Gutierrez (BSC, ESP) Jiri Vondrasek (ELIXIR, CZ) Anna Strachotova (ELIXIR, CZ) Marcelo Martí (FCEN-UBA, AR) Esteban Mocksos (FCEN-UBA, AR)
13:30 - 14:30	***Lunch Break***
Session 6: IDPfun2 Consortium	
Chair: Miguel Andrade	
14:30 - 15:00	Invited Speaker: ***Gabor Erdos*** *(ELTE, Hungary)* *A zero-shot thermodynamic potential for protein mutational stability*
15:00 - 15:30	Invited Speaker: ***Nathalie Sibille*** *(CNRS, France)* *Functional disorder in cell signaling: molecular mechanisms and case studies*

15:30 - 16:15 *Short talks selected from abstracts (8'+2' questions)*	**Juan Mac Donagh** *(UNQ, BA, Argentina)* *Transposable elements as evolutionary substrates of protein disorder in the human proteome*
	Heli Garcia Alvarez *(UNSAM, BA, Argentina)* *LINKER-Pred: A Deep Learning Method and Web-Server for the Prediction of Disordered Flexible Linkers in Proteins*
	Leandro A. Bugnon *(sinc, UNL StaFe, Argentina)* *PonTED: augmenting flexible-linker prediction with protein language models and inter-domain context*
	Marina Navarro *(UNQ, BA, Argentina)* *Structural Diversity and Conformational Plasticity of Plantaricin A Variants*
16:15 - 16:30	***Coffee Break***
16:30 - 18:00	Poster Session
18:00 - 18:45	Closing Keynote Talk: ***Salvador Ventura*** *(UAB, Spain)* *Targeting protein aggregation at the interface of experiment and computation*
18:45 - 19:15	**Prizes and Closing Ceremony**
19:15 - 19:30	**A2B2C meeting - Election & change of Authorities**